

Total no. of Pages 2

Roll No.....

THIRD SEMESTER- B. Tech.
END-SEMESTER EXAMINATION, November, 2024

Corse Code: CGECC302

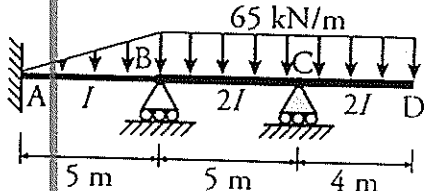
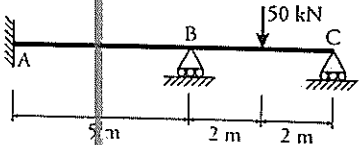
Corse Title: Signal and System for Geoinformatics

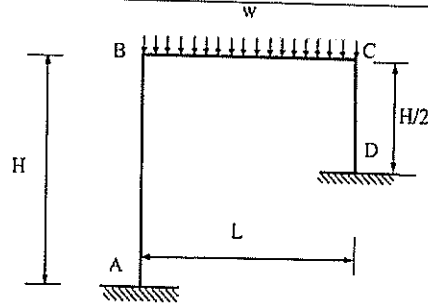
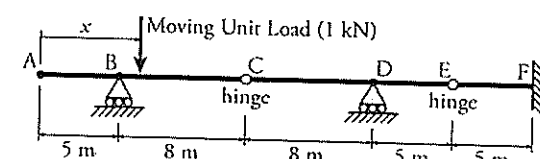
Time: 3Hr

Max. Marks: 50

Note: Attempt all questions. Missing data/information (if any), may be suitable assumed & mentioned in the answer.

Q. No.	Question	Marks	CO
Q 1	Attempt any 2 parts of the following		
1a	What are discrete time signals-impulse signal, unit signal and ramp signal?	5	CO1
1b	What are (i) Energy and Power signals (ii) Even and Odd signals (iii) Periodic and Non Periodic signals (iv) Analog and Digital signals?	5	CO1
1c	What are (i) Linear and Non Linear systems (ii) Causal and Non Causal systems (iii) Time variant and Time Invariant systems (iv) Stable systems?	5	CO1
Q 2	Attempt any 2 parts of the following		
2a	Compute the convolution of $x(n) = 2^n u(n)$ and $h(n) = \delta(n-1) - 7\delta(n-2)$	5	CO2
2b	Find $y(n)$ if $h[n] = \begin{cases} 3, & n = \pm 1 \\ 1, & n = 0 \\ 0, & \text{otherwise} \end{cases} \text{ and } x[n] = \begin{cases} 1, & n = 0 \\ 4, & n = 1 \\ -5, & n = 2 \\ 0, & \text{otherwise} \end{cases}$	5	CO2
2c	What are the properties of discrete time Fourier series?	5	CO2
Q 3	Attempt any 2 parts of the following		

Q 3	Attempt any 2 parts of the following	8	
3a	A three hinged parabolic arch of span 'l' meter and central rise 'h' carries a load w per unit length over the left half of the span. Compute the position and magnitude of maximum positive bending moment?	4	3
3b	Derive the expression for strain energy in axial loading and in bending?	4	3
3c	Explain Castigliano Theorem?	4	3
Q 4	Attempt any 2 parts of the following	8	
4a	Analyse the given figure using Slope deflection Method? 	4	4
4b	Draw BMD and SFD of the given figure using Moment Distribution Method? 	4	4
4c	Analyse the figure using any iterative approach or slope deflection equation.	4	4

			
Q 5	Attempt any 2 parts of the following	8	
5a	Draw ILD for vertical reaction at B, D and F using analytical method or Muller Breslau's principle? 	4	5
5b	Derive the ILD ordinates for vertical reactions at both ends, shear force and bending moment at section 'x' in a overhanging beam carrying a unit point load? Also, give the values of ILD ordinates.	4	5
5c	A simply supported beam AB of span 22 m carries a uniformly distributed load of 12 kN/m magnitude. The uniformly distributed load is of 12 m in length. Determine the maximum bending moment at section 14 m from right support A?	4	5